Modelling Nonlinear Relationships in Financial Markets: VIX vs High-Yield Spreads

Objective

- **Demonstrate** a **practical workflow** for the **diagnosis** of **non-linear relationships** in financial data:
  - Detection
  - Modelling
  - Diagnosis

- Apply multiple functional forms and diagnostics

- In this example it is **NOT the objective** of “picking a winner” purely by RMSE

Data

VIX

- Market **implied volatility** of the S&P 500 over ~30 days (options-derived)
- Common **risk-off / fear gauge**: higher VIX ⇒ higher perceived equity risk

HY OAS (BAMLH0A0HYM2)

- **Option-adjusted spread** of a broad US high-yield corporate bond index
- Spread over the **US Treasury curve**, expressed in bps (credit + liquidity premium)

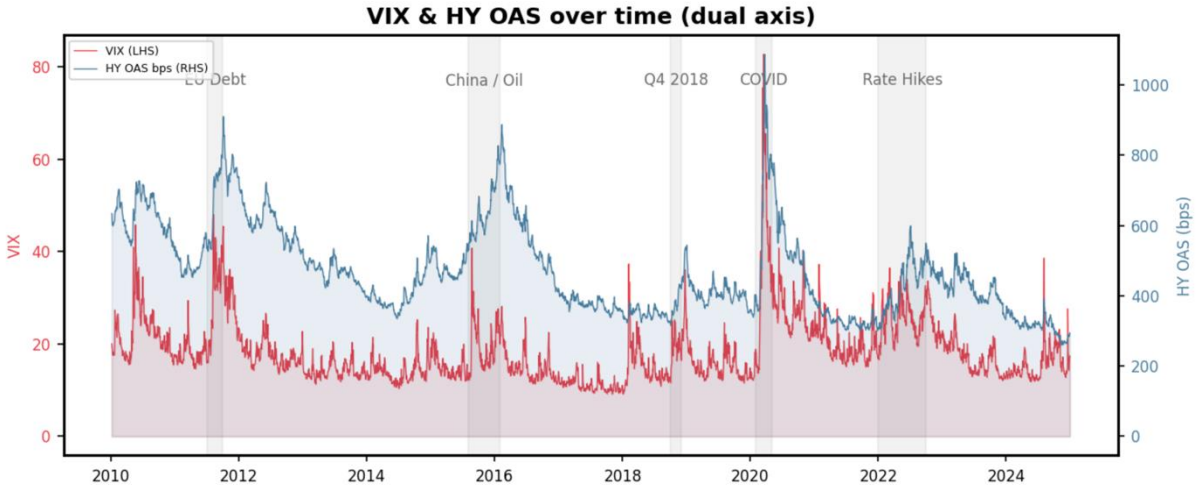
Relationship (why we expect nonlinearity)

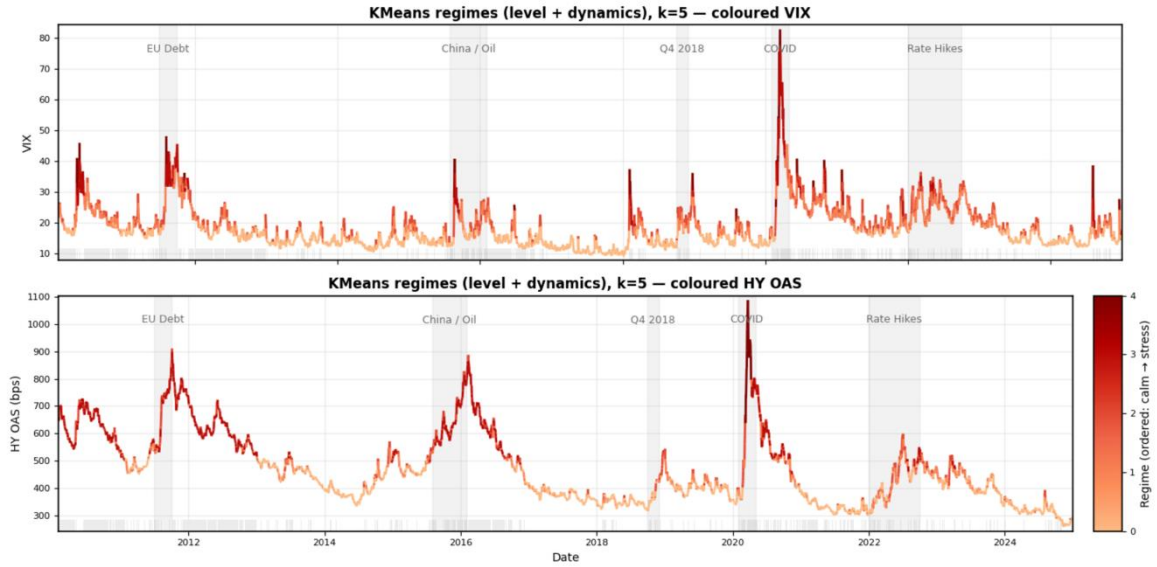
- In **risk-off** episodes: **VIX rises**, **HY OAS widens** (spreads increase), risk assets weaken (e.g., SPY down)
- The mapping is typically **state-dependent**: mild response in calm markets, sharper/convex response in stress regimes (liquidity + risk aversion feedback)
  - relationship is rarely linear: mild moves at low stress, then **convex** responses under stress

Risk-off is a market regime where investors reduce exposure to risky assets and shift toward safer, more liquid holdings

Data

- **VIX** in points (e.g., 15–30 normally, spikes in crises)
- **HY OAS** (BofA/ICE series) converted from percent to **basis points**





Top Chart

- Shared shock timing, different persistence
- Major episodes (EU debt, China/oil, Q4'18, COVID, rate hikes) show up in **both** series
- VIX reacts fast and mean-reverts quickly
- **OAS is slower and "sticky"** (stress persists after the volatility spike)

Bottom Chart

- KMeans (k=5, dynamics) labels market **states calm / stress**
- Stress regimes align strongly in big crises, but **duration differs**: OAS stays stressed longer than VIX

Relationship is **regime-dependent**: can't assume one stable linear mapping from VIX to OAS across all market states

Data

Raw levels

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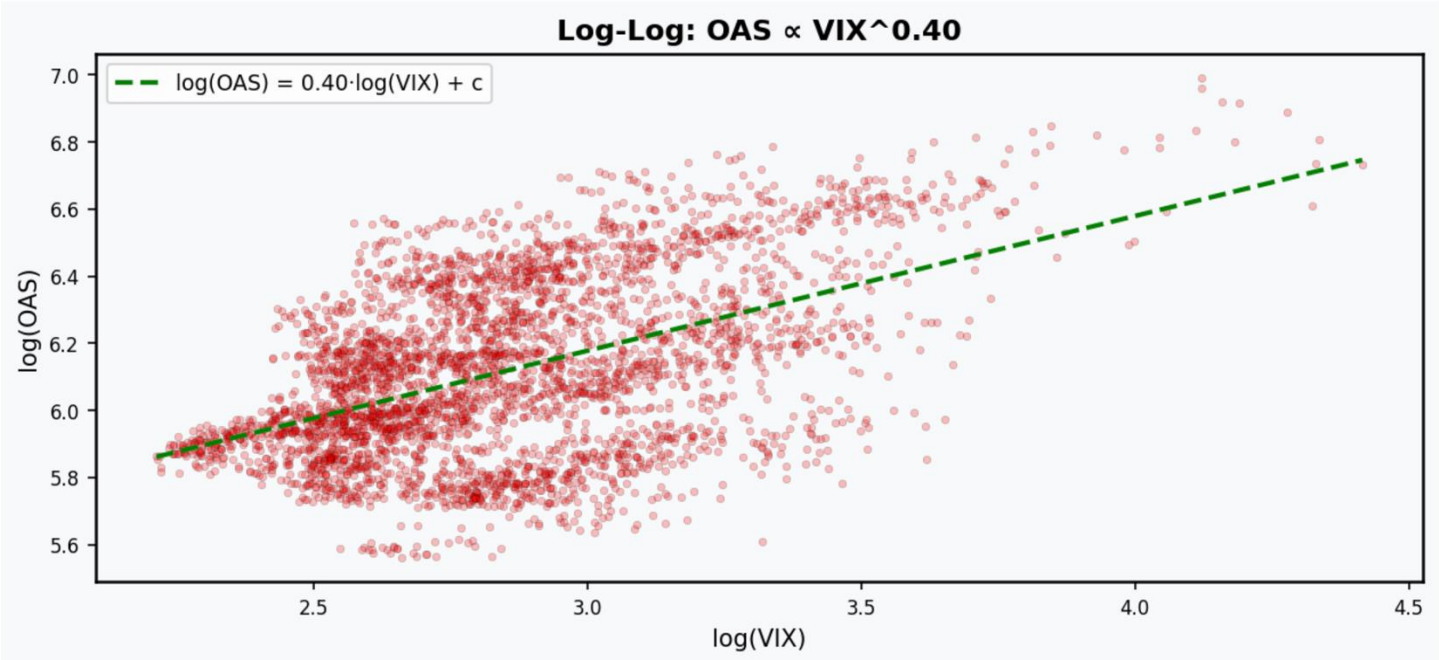

The relationship is not well-captured by a single straight line in raw levels

Transformation:

1. log_VIX = log(VIX)
2. log_OAS = log(OAS_bps)

$$\log(OAS) = \alpha + \beta \log(VIX) \Rightarrow OAS \propto VIX^{\beta}$$

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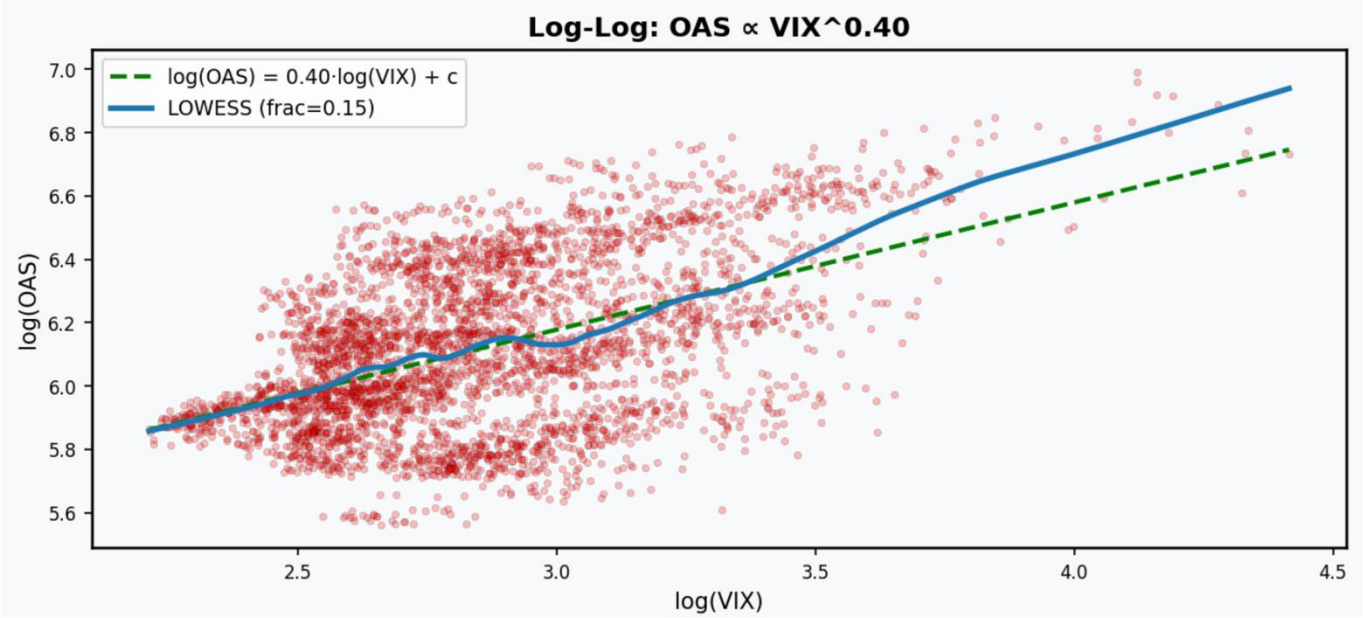
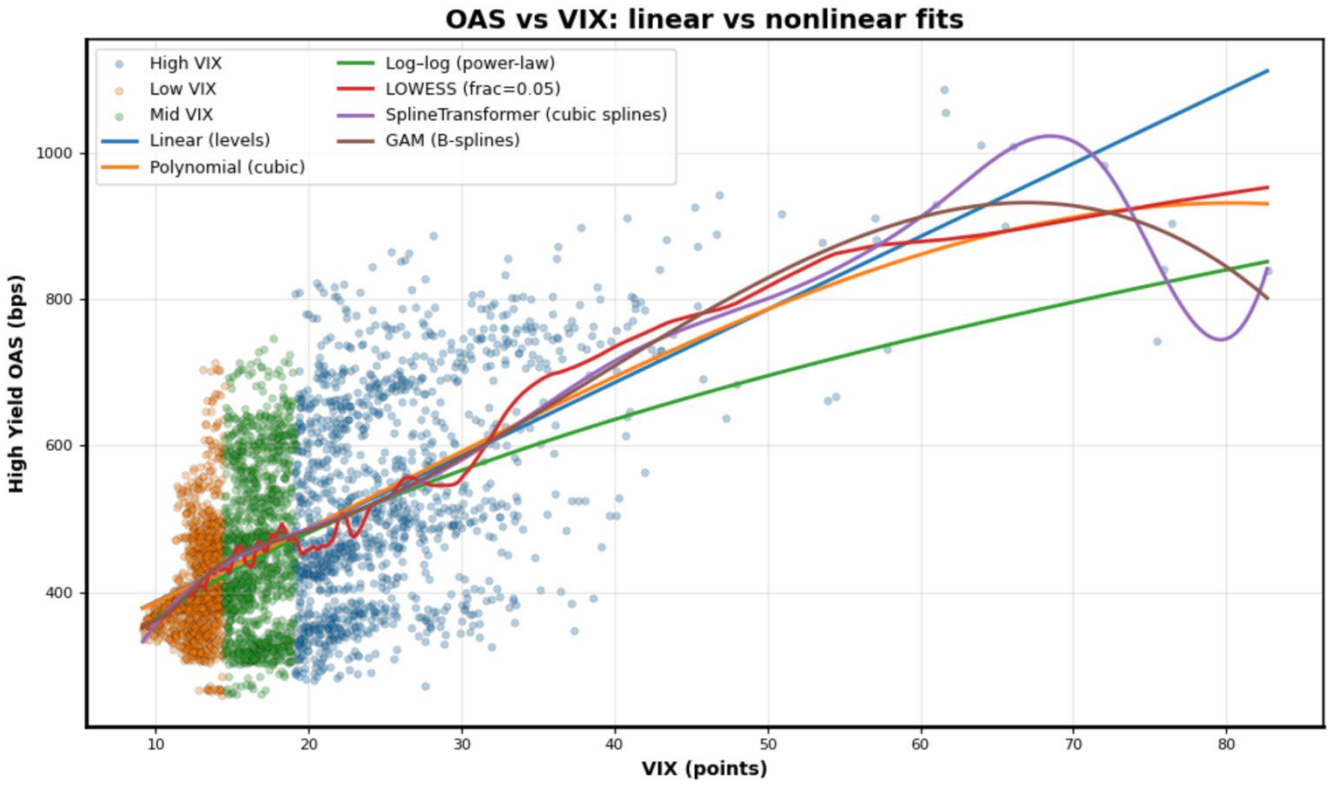


- **Positive relationship, but not perfectly linear in log–log space : S-shape curvature**
- Flatter / wavy in the mid-region (around log(VIX) ≈ 2.8–3.2)
- **Steepens materially at higher log(VIX)** (roughly > 3.3–3.5) - **Implies Convexity**
  - When volatility is already high, additional increases in VIX correspond to **disproportionately**
- **Single beta (0.40) is a good baseline, but the LOWESS suggests you'll improve realism with nonlinear structure**

Models

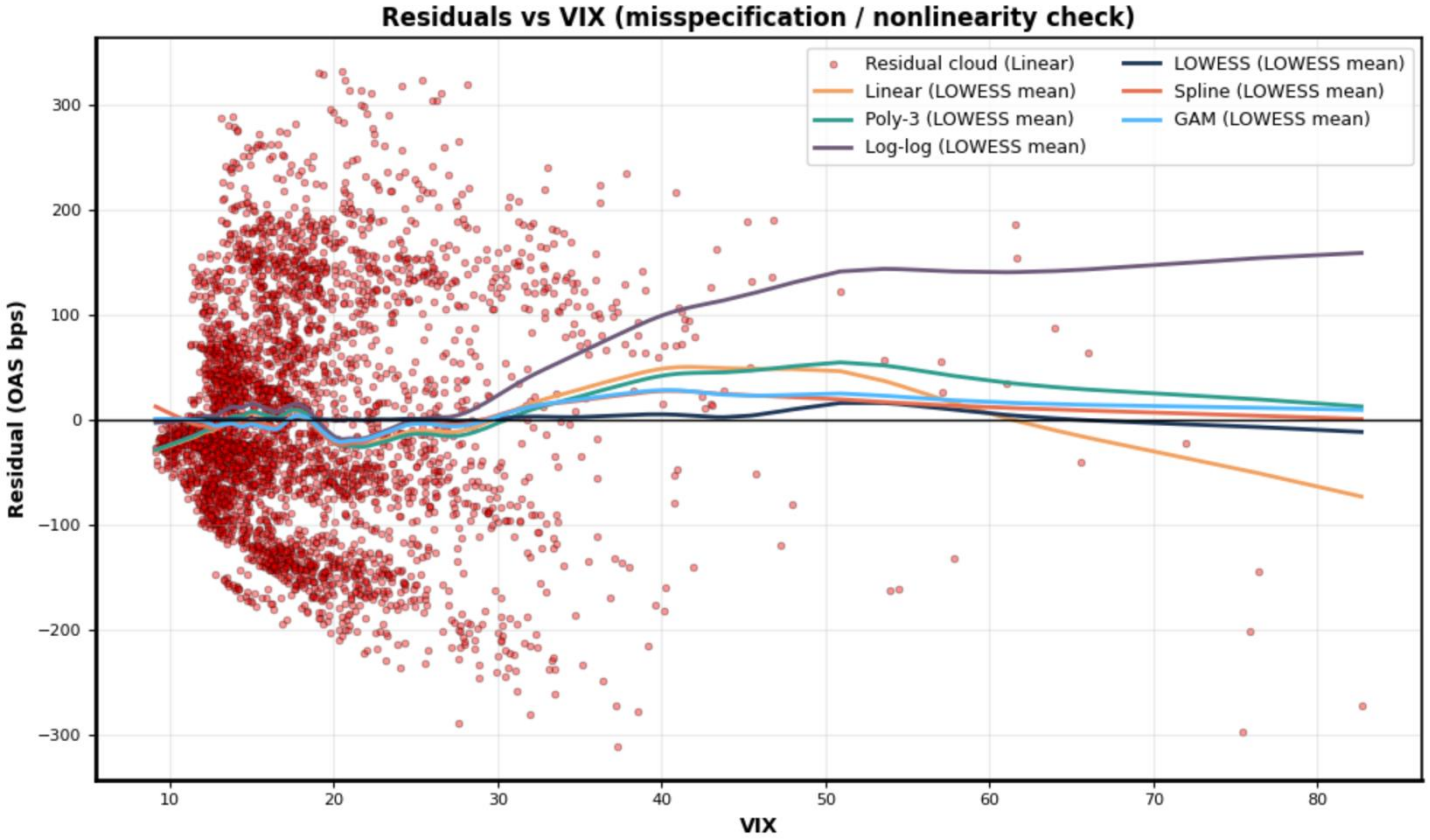
1. **Linear (levels)**: $OAS = a + b\,VIX$
2. **Polynomial (cubic)**: $OAS = a + b_1 V + b_2 V^2 + b_3 V^3$
3. **Log–log (power law)**: $\log(OAS) = a + \beta \log(VIX)$
4. **LOWESS** (nonparametric smoother)
5. **SplineTransformer**
6. **GAM with B-Splines**

Multiple Fits – Regime Colouring VIX Volatility (Low, Mid, High)



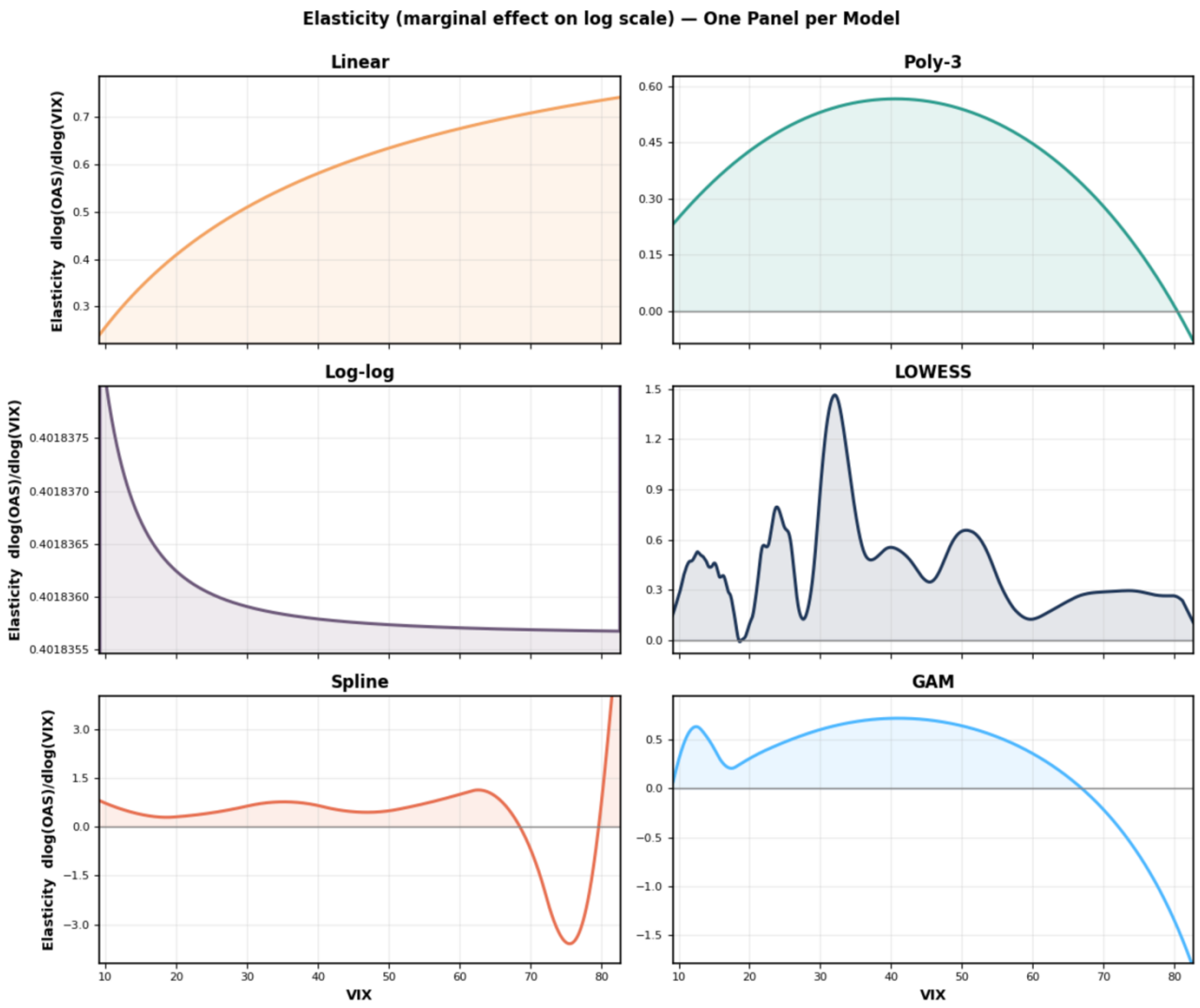
- There in no one stable OAS = f(VIX) relationship, high VIX dispersion widens significantly: **state dependent volatility**
- **VIX** is a **good** indicator of **market stress**, but in **high-stress** periods **OAS becomes much harder to predict** from **VIX alone**
- Variety of models basically inidicate: **monotone up (keeps rising** or **flattens)**, **nonlinear**, **tail uncertainty**
- **Conclusion: Regime-conditional model or similar…**

Residuals - Strong Heteroskedasticity and Structure



- Residual dispersion is **very wide** at low-to-mid VIX and remains wide into high VIX
- For conditional mean $\mathrm{E}[OAS \mid VIX]$, the **average residual** should be ~0 at every VIX level
- **Log–log (purple) is far above zero at moderate/high VIX**
- **Mean vs Variance:** models reduce mean-bias but the residuals shows large, regime-dependent variance
- **Regime dependence is real**
- Strong heteroskedasticity in risk-off regimes

Elasticity – E(*x*) versus VIX



- **Two concepts**
  - Marginal Effect (marginal effect in levels) $ME(x) \approx \frac{\Delta OAS}{\Delta VIX}$
    - Absolute Sensitivity
  - **Elasticity (log-scale marginal effect)** $E(x) \approx \frac{\%\Delta OAS}{\%\Delta VIX}$
    - Normalized sensitivity
- **Linear - Elasticity rises with VIX**
- **Poly-3 – Elasticity increases into mid VIX then declines (high VIX)**
- Log-Log : Elasticity collapses to the slope coefficient $\beta = here\ 0.4$
  - Does not represent regime variation
- Lowess : Elasticity varies substantially with VIX
  - Regime dependency?
- Spline : becomes unstable, negative in the far-right tail
- GAM : smooth mid-range behaviour

- Use the **log–log elasticity (~0.4)** as the reliable headline figure
- **LOWESS/GAM/Spline -** signs that the VIX–OAS linkage is **regime-dependent** i.e. sensitivity changes across market states
- **Mid-range variation** might indicate meaningful "regime effect"

Summary

1. VIX and OAS have a real relationship, but it is not uniform

2. The two series have different dynamics
   - VIX is spiky and mean reverting
   - OAS – Slower and more sticky

3. Flexible non-linear models detect structure but unstable in the tails

4. LOWESS / spline / GAM highlight **local curvature** and **transition regimes – suitable for diagnostics**

5. Log–log provides a **stable, interpretable, unit-free elasticity**